

NON-TECHNICAL DRIVERS AND BARRIERS RELATED TO ENERGY FLEXIBILITY FOR CONSUMERS AND INDUSTRY

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SOCIAL INNOVATION

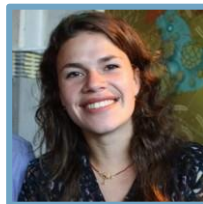
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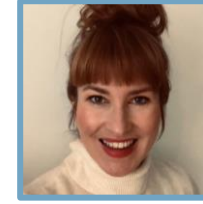
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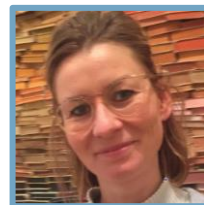
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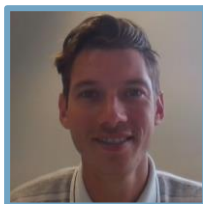
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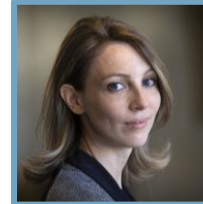
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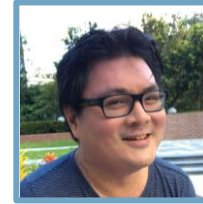
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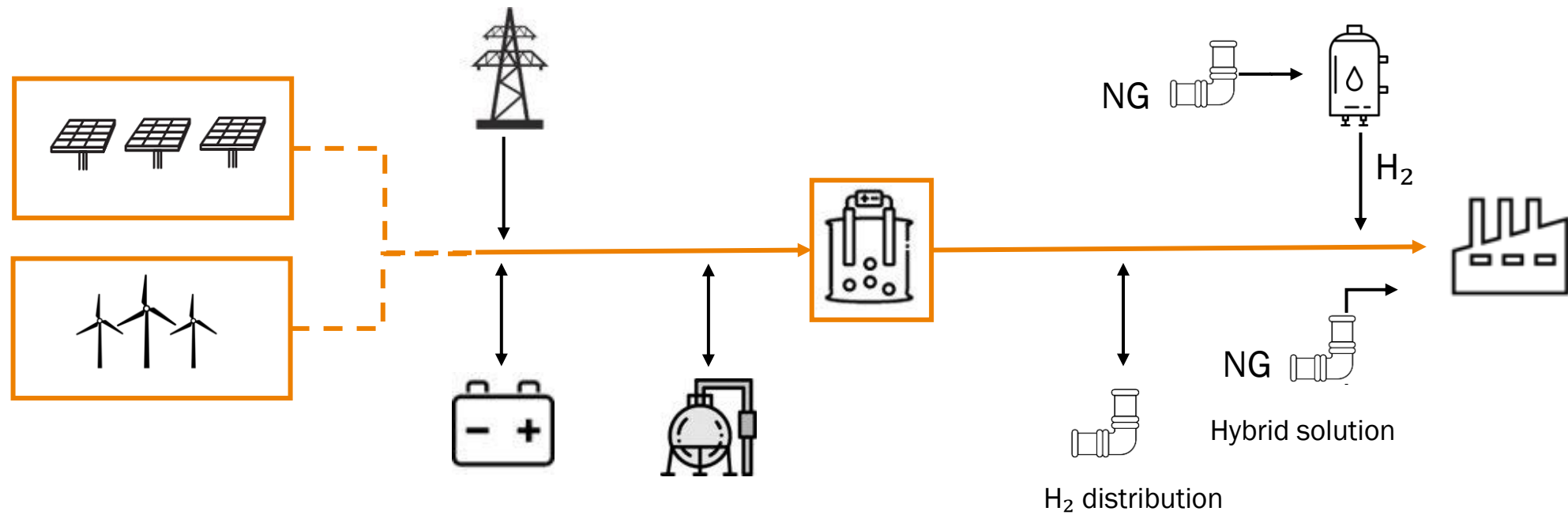


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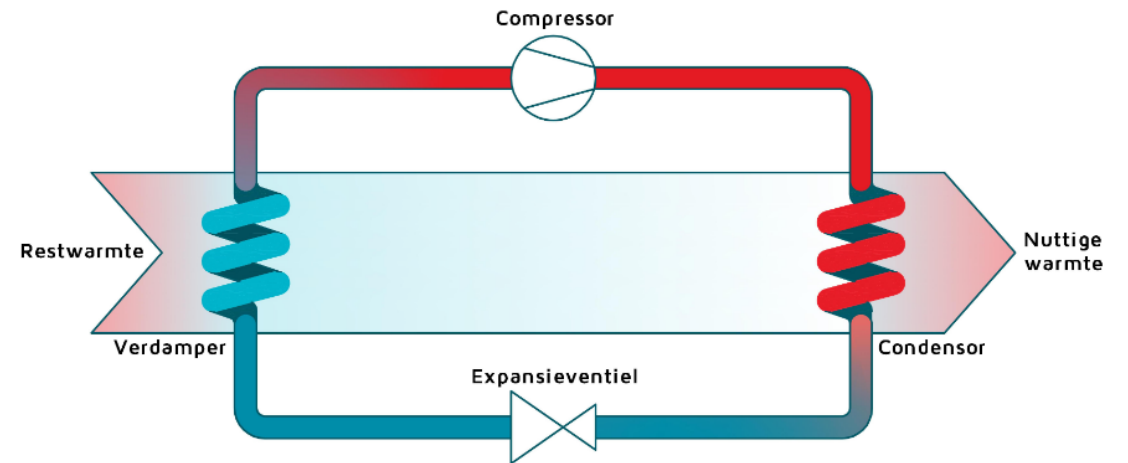


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› OPTIONS FOR PROVIDING FLEXIBILITY



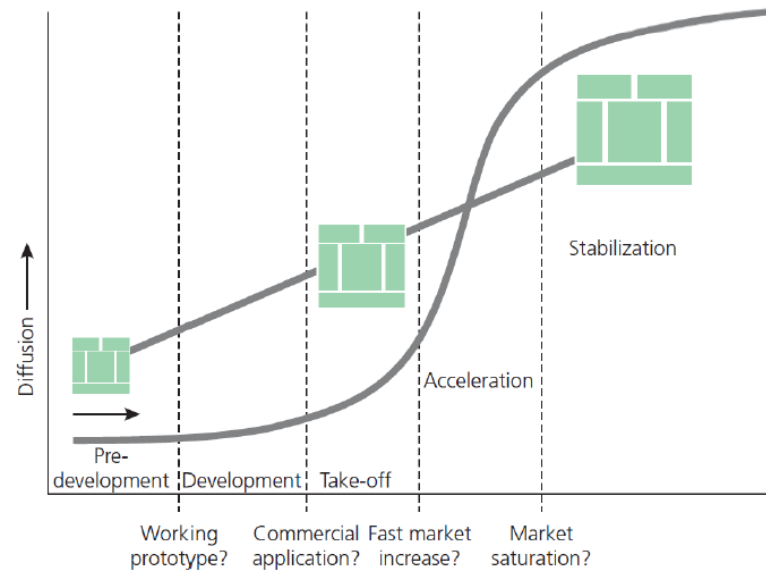
› INDUSTRIAL HEAT PUMPS



› **DISCUSSION TOPIC**

- › Industrial heat pumps have clear advantages over conventional technologies (gas-fired steam boilers).
- › Technological concept and components are proven.
- › Yet, not large-scale rollout of industrial heat pumps has taken place.
- › What are reasons for the stalled market implementation and roll-out of industrial heat pumps?
- › Could current developments (gas prices, independence from russian gas) give an impetus for implementation?

› BARRIERS TO IMPLEMENTATION HEAT PUMPS



› THE KICKSTART PROJECT

› Market creation:

- › How can innovators & early adopters involve other stakeholders and foster large-scale rollout?
- › How can the relationships between suppliers and end-users of industrial heat pumps be improved?

› Willingness to invest:

- › What barriers (e.g. lack of knowledge, complex decision making, resistance to change) within the end-user prevent them from investing?
- › Do these barriers differ between types of end-users?



› For more information: [Kickstart market industrial heat pumps - Topsector Energie](#)

› THE KICKSTART PROJECT

Method:

- › Study 1: Case study research at four end-users.
- › Study 2: Workshop series to identify leverage points for change in the innovation ecosystem.
- › Study 3: Develop tool for assessing economic viability of industrial heat pumps, incl. dynamic response & flexible operation (Recoy)
- › Study 4: Develop recommendations for accelerating market implementation

› SMART ENERGY SERVICES – CONSUMER PERSPECTIVE

Projects

- GO-E (Gebouwde Omgeving Elektrificatie)
[GO-e onderzoekt energieflexibiliteit in gebouwde omgeving | TNO](#)
- Bright (Boosting DR through increased community-level consumer engaGement by combining Data-driven and blockcHain technology Tools with social science approaches and multivalue service design
[Bright Project](#)



› GO-E (GEBOUWDE OMGEVING ELEKTRIFICATIE)



- › Can smart flexibility services provide an alternative to grid reinforcement in the built environment?
- › GO-E consortium: Alliander, Enexis, Stedin, ElaadNL, Greenchoice, Recoy, Itho Daalderop, ETPA, Technolution, Phase to Phase, DNV GL, Witteveen + Bos, TU Eindhoven, TU Delft, coordinated by TNO
- › Three living labs: Houten, Heeten and Loenen (Veluwe)
- › Focus on Dutch citizens (not only front runners)



› VALUE-BASED DESIGN

- › Value-based design: exploring the core psychological values that are important to people, the real drivers or motivators of people. From there, we can design smart energy services that correctly address these drivers/motivators.

- › Values of Sheldon (2001)

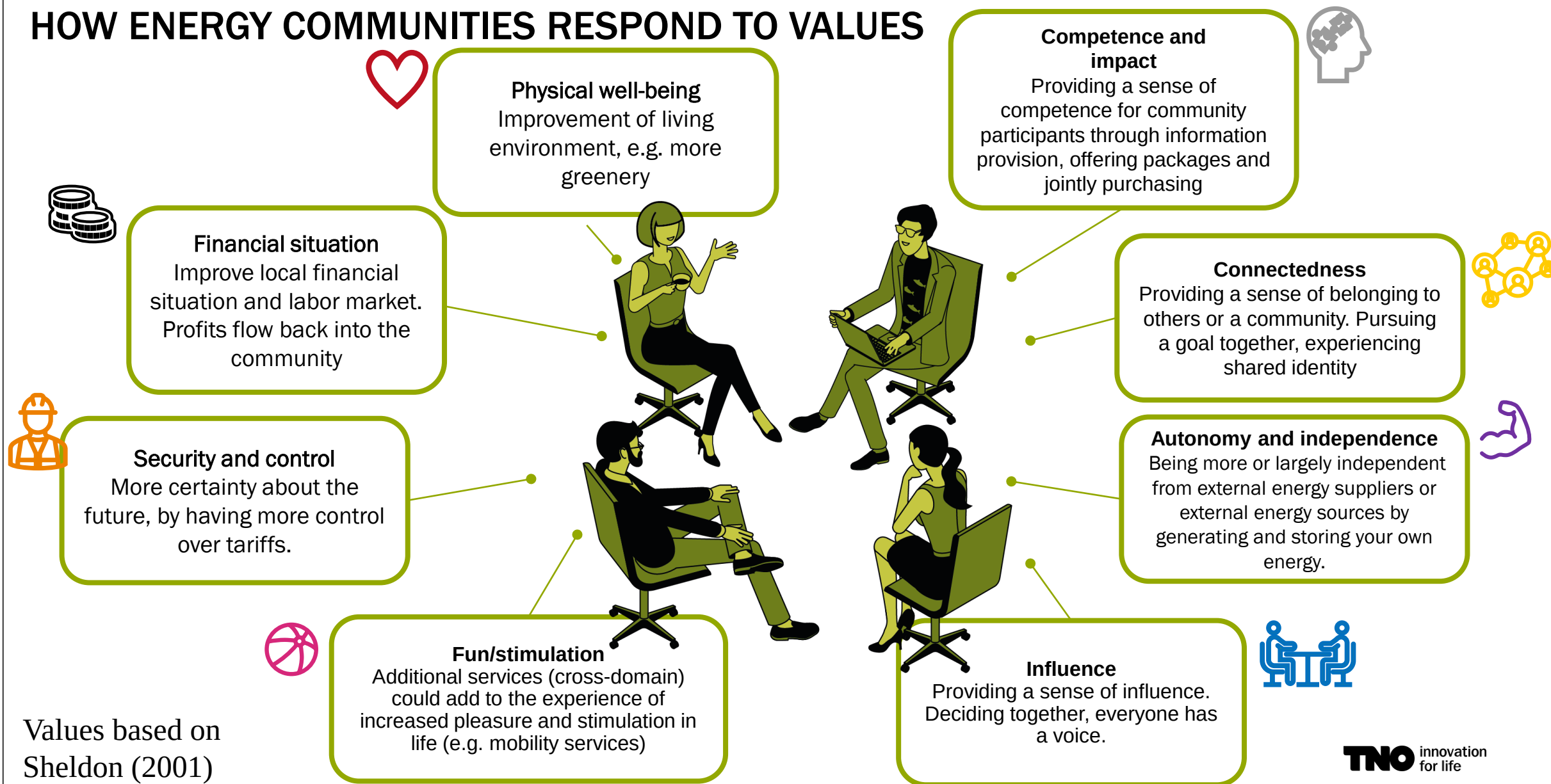


- › Attention to people in roles of citizens and consumers (citizen and consumer engagement)
- › Away from the usual financial incentives, more towards what citizens and (groups of) consumers really care about.
- › Role of energy communities

Sheldon KM, Elliot AJ, Kim Y, Kasser T. (2001). What Is Satisfying About Satisfying Events? Testing 10 Candidate Psychological Needs. *Journal of Personality and Social Psychology*. 2001;80(2):15. doi:10.1037//0022-3514.80.2.325

VALUE-BASED DESIGN

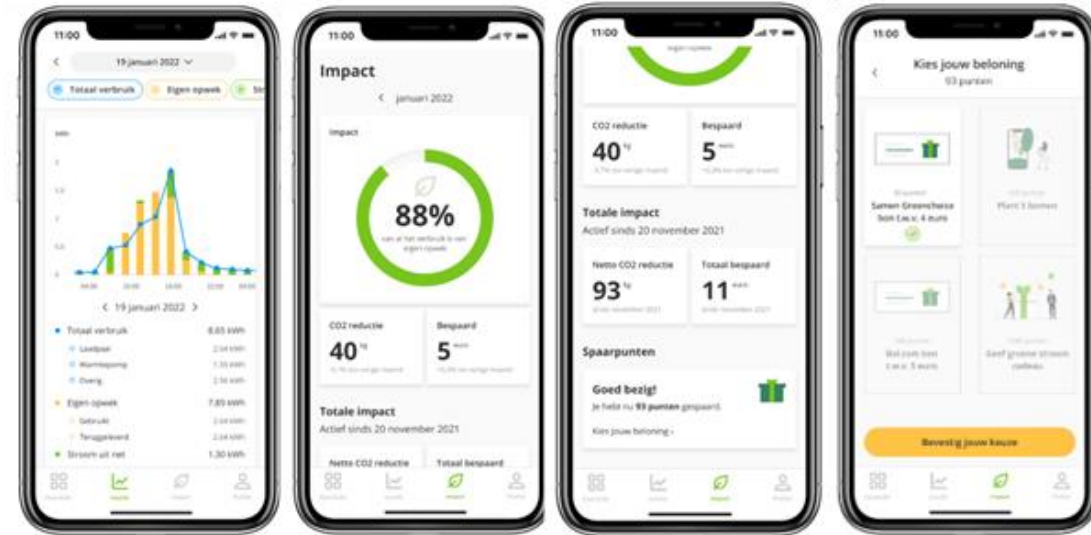
HOW ENERGY COMMUNITIES RESPOND TO VALUES



GO-E

UNLOCKING FLEXIBILITY

- Research question: How much flexibility is there to unlock from a consumer perspective with PV, heat pumps, EV and home batteries?
- Research methods: interviews, co-creation sessions
- **Conclusion:** in this specific living lab Houten (Dark Green segment)
 - no interest in financial incentives
 - interested in a number of core values such as autonomy, competence in putting one's own intrinsic motivation into practice, connectedness and physical and mental well-being.

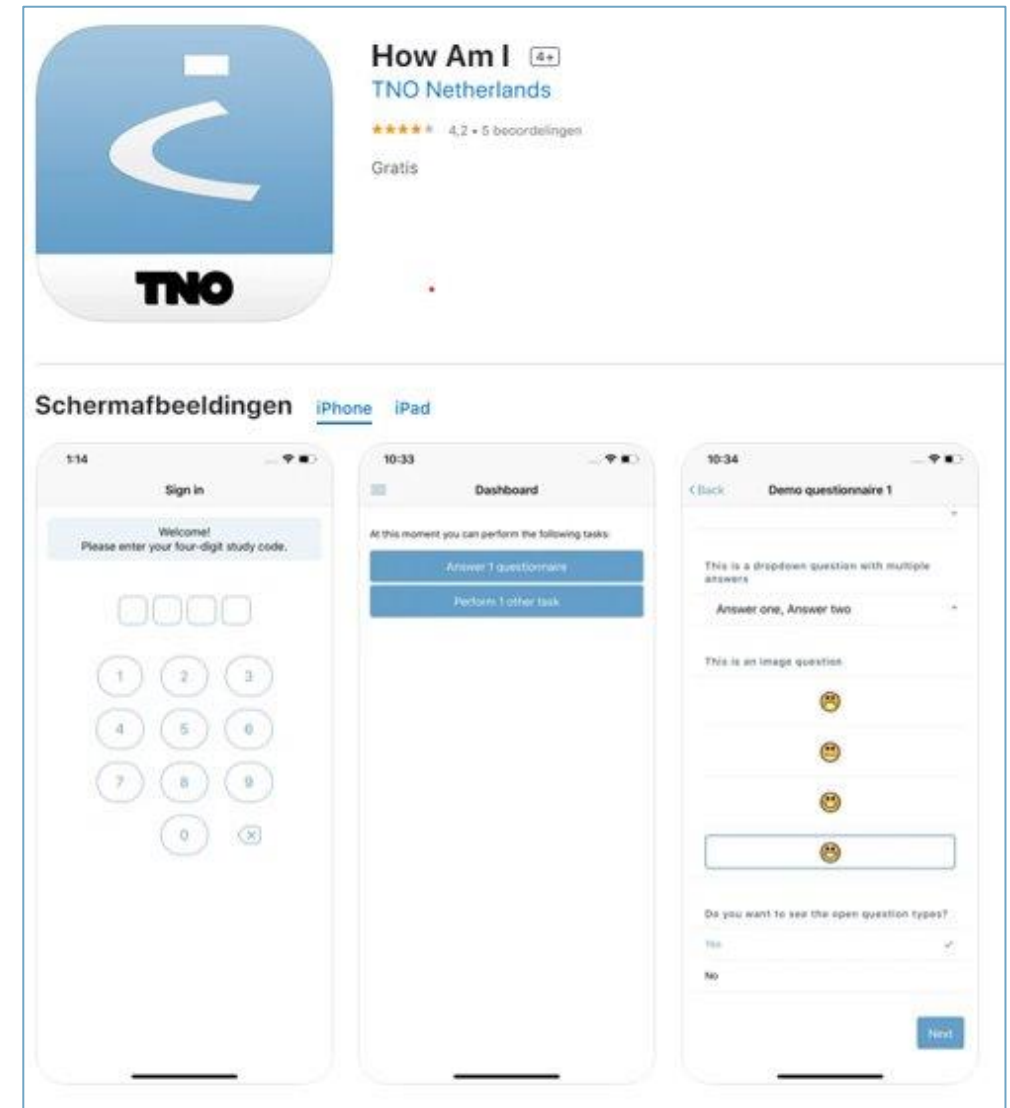


Living lab Houten (designs by Angi Studio)



NEXT STEP: EXPERIENCE SAMPLING

- Experience sampling: measuring user experiences and desires in everyday life (real time)
- Define how much flex can be offered
- Evaluation of the flex mechanisms on user experience



› **THANK YOU FOR YOUR ATTENTION**

Projects

Kickstart

GO-e onderzoekt energieflexibiliteit in gebouwde omgeving | TNO

Bright Project

BRIGHT_D3.1_TNO.pdf (brightproject.eu) - Overview of barriers and drivers for consumer engagement in DR

Newsbrief (Dutch)

Nieuwsbrief inschrijving - Energy.nl

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